MIL-H-27269A (USAF)
AMENDMENT 2
28 AUGUST 1981
Superseding
AMENDMENT 1
1 November 1978

MILITARY SPECIFICATION

HORIZONTAL SITUATION INDICATOR GROUP AF/A245J-1

This amendment forms a part of Military Specification, MIL-H-27269A (USAF), dated 20 December 1977, and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1: Under "SPECIFICATIONS, Federal", delete "QQ-P-416".

PAGE 2

2.1: Under "SPECIFICATIONS, Military", delete "MIL-E-5272" and add:

"MIL-C-83488	Coating, Aluminum, Ion Vapor Deposited"

Under "STANDARDS, Military" change "MIL-STD-100" to read "DoD-STD-100". Add the following:

"MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods"

PAGE 3

3.1: Delete in its entirety and substitute:

"3.1 First article. When specified (see 6.3), the contractor shall furnish sample units for first article inspection and test (see 4.5)."

PAGE 5

3.6,e: Delete "Spray" and substitute "Fog".

PAGE 6

3.6,o: Add new sub-paragraph:

"o. Mean-time-between-failures (MTBF). The horizontal situation indicator group shall have a MTBF of not less than 775 hours (0_o)

FSC 6610

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at a confidence factor of 90 percent and discrimination ratio of 2. Testing to verify this requirement shall be in accordance with MIL-STD-781C, Test Plan III C (see 4.4.24.2)."

PAGE 11

3.10; line 5: Delete "MIL-E-5272" and substitute "MIL-STD-810".

PAGE 16

Figure 5: Delete Figure 5 and substitute new Figure 5 attached in this amendment.

PAGE 27

3.29.2: Delete paragraph and substitute new paragraph as follows:

"3.29.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

PAGE 28

Add new paragraph 3.33:

"3.33 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item."

PAGE 29

4.1, b: Delete "Qualification" and substitute "First article".

PAGE 31

4.2.2.2, h: Delete "Salt spray" and substitute "Salt fog".

4.2.2.2.1, Group II: Delete "Salt Spray" and substitute "Salt fog".

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4.4.15: Delete existing paragraph and substitute the following:

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"4.4.15 Vibration error. The indicator shall be mounted, in its normal operating position, on a vibration fixture, and properly connected for operation. While being operated, the indicator shall be vibrated with circular motion in a plane inclined 45 degrees to the horizontal with a total vibration amplitude between 0.003 and 0.005 inch. The frequency of applied vibration shall be varied slowly from 5 to 50 cycles per second. While being vibrated, the indicator shall meet the following requirements."

PAGE 38

4.4.22, line 3: Delete "MIL-E-5272" and substitute "MIL-STD-810".

4.4.22.1, line 1: Add Method 501.1" before "Procedure I".

4.4.22.2: Delete entire paragraph and substitute the following:

"4.4.22.2 Temperature altitude. The horizontal situation indicator group shall be subjected to a temperature-altitude test in accordance with Method 504.1 of MIL-STD-810 for equipment category 6. There shall be no binding or sticking at any possible indication. After the horizontal situation indicator group has returned to room temperature, it shall meet all individual tests. There shall be a 5-minute warmup period after the application of power before data is taken."

PAGE 39

4.4.22.3: Add "Method 503.1" before "Procedure I".

4.4.22.4: Add "Method 507.1" before "Procedure I".

4.4.22.5: Add "Method 508.1" before "Procedure I".

4.4.22.6, title: Delete "Salt Spray" and substitute "Salt fog". Add "Method 509.1" before "Procedure I" in line 1.

4.4.22.7, title: Delete "Sand and Dust" and substitute "Dust (fine sand)". Add "Method 510.1" before "Procedure I" in line 1.

4.4.22.8: Delete entire paragraph and substitute the following:

"4.4.22.8 Vibration failure. The indicator shall be subjected to a vibration type failure test as follows: The indicator shall be mounted on the vibration apparatus in its normal operating position. It shall be properly interconnected for operation. While being operated, it shall be vibrated with a circular motion of 0.018 to 0.020 inch diameter in a plane inclined 45 degrees to the horizontal plane and the frequency of vibration shall be varied uniformly form 5 to 50 cycles per second and return once

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each hour for a 3 hour period. Following the vibration test, the horizontal situation indicator group shall meet the requirements as specified in 4.4.4 through 4.4.8. No looseness in the mechanism nor damage to any part of the horizontal situation indicator group shall result from the vibration."

PAGE 40

Add new paragraph 4.4.24.2 as follows:

"4.4.24.2 Mean-time-between-failure (MTBF). Equipment on-time accumulated during sampling/life tests shall be used in computation of MTBF. Accept/reject criteria shall be as specified in MIL-STD-781, Test Plan III C."

PAGE 41

4.5 through 4.5.3: Delete "Qualification" where mentioned and substitute "First article".

PAGE 43

6.3: Delete entire paragraph and substitute the following:

"6.3 First article. A first article is required for inspection and it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Aquisition Regulation. First article tests are described in 4.5. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article."

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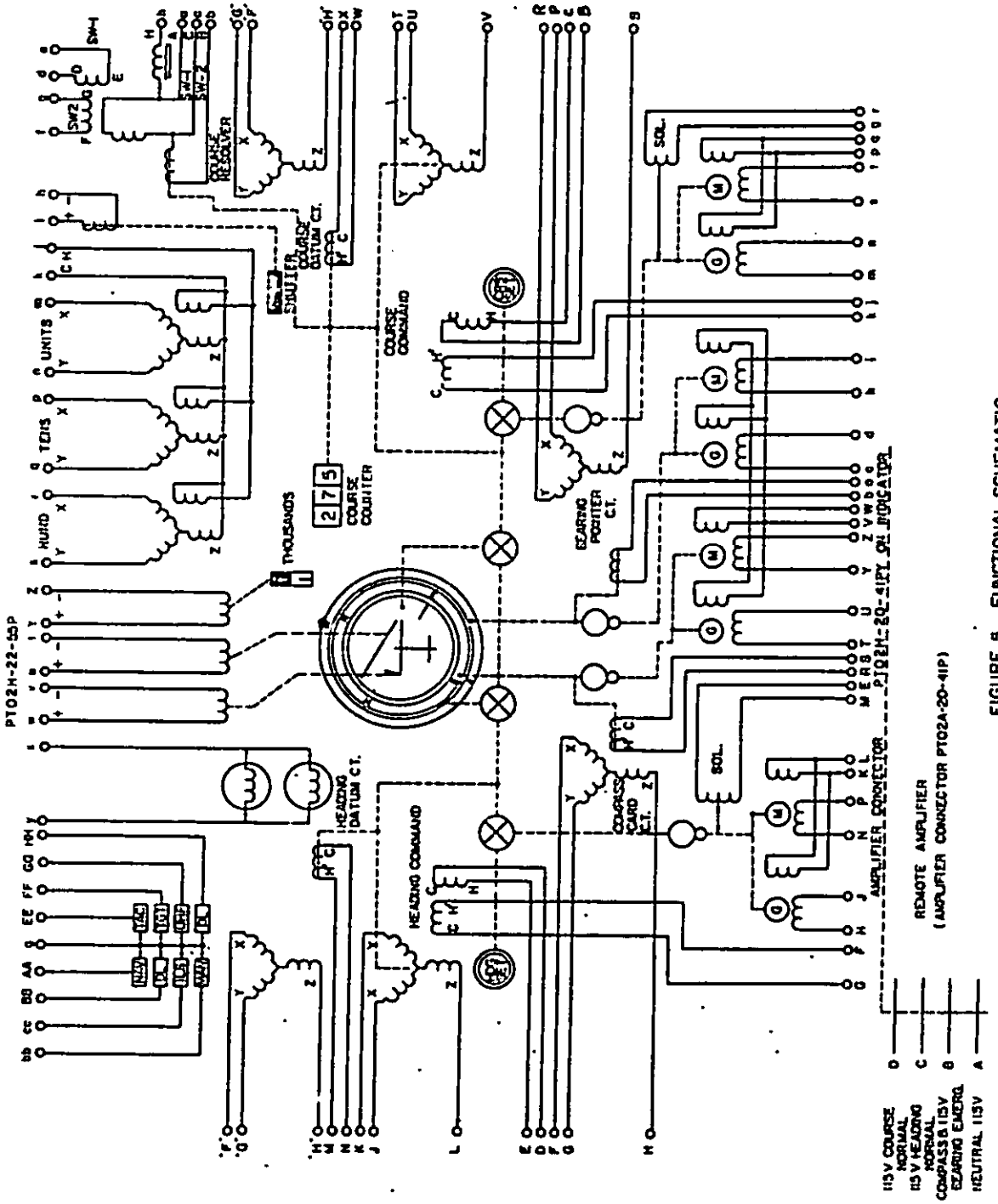


FIGURE 5. FUNCTIONAL SCHEMATIC

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